

CHE 435/525 *Process Systems Analysis and Control*

## Lecture 4: Laplace Transform and Its Use in ODEs

Wentao Tang

Assistant Professor, Chemical & Biomolecular Engineering  
North Carolina State University

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# Discuss & Recapture

- What complicates the modeling when the system involves reactions?



- How do we account for multiple reactions?
- \*\* What behaviors can a nonlinear system have?

# Learning objectives of this lecture

- ① Calculate Laplace transform of common functions of time
- ② Calculate inverse Laplace transforms through **partial fraction expansion**
- ③ Analytically calculate the output response to input signals

## A quote from Laplace

*Ce que nous connaissons est peu de chose, ce que nous ignorons est immense.* (What we know is not much. What we do not know is immense.)

Pierre-Simon, Marquis de Laplace (1749–1827)

# Laplace transform

## Definition

Given function (“signal”)  $f(t)$  on  $t \geq 0$ , its Laplace transform is

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t)e^{-st} dt \quad (1)$$

comprehended as a function of a **complex variable**  $s$ .

- Property 0: Laplace transform is linear:

$$\mathcal{L}[af(t) + bg(t)] = a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)]$$

- Example 0: Step function

$$S(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \Rightarrow \mathcal{L}[S(t)] = \int_0^{\infty} 1e^{-st} dt = \frac{1}{s}.$$

For a step of magnitude  $a$ :

$$\mathcal{L}[aS(t)] = a/s$$

# Exponential and sinusoidal functions

- Property 1: Exponential term

$$\mathcal{L}[f(t)e^{at}] = \int_0^{\infty} f(t)e^{at-st} dt = F(s-a).$$

- Example 1: Exponential function

$$\mathcal{L}[e^{at}] = \frac{1}{s-a}.$$

- Sine and cosine:

$$\begin{aligned}\mathcal{L}[\cos at] &= \mathcal{L}\left[\frac{e^{iat} + e^{-iat}}{2}\right] = \frac{1}{2} \left( \frac{1}{s-ia} + \frac{1}{s+ia} \right) = \frac{s}{s^2 + a^2}. \\ \mathcal{L}[\sin at] &= \mathcal{L}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] = \frac{1}{2i} \left( \frac{1}{s-ia} - \frac{1}{s+ia} \right) = \frac{a}{s^2 + a^2}.\end{aligned}$$

# Differentiating/Integrating over time

- Property 2: If  $f$  is differentiable,

$$\mathcal{L}[f'(t)] = sF(s) - f(0)$$

- ▶ If  $f(0) = 0$ ,  $\mathcal{L}[f'] = sF(s)$  ( $d/dt \Rightarrow$  multiplication by  $s$ )
- ▶ More generally,

$$\mathcal{L}[f^{(n)}] = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0).$$

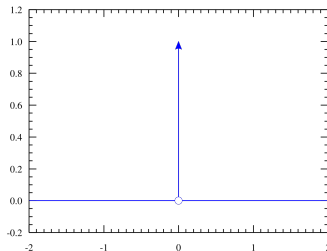
- Hence, if  $g(t) = \int_0^t f(\tau)d\tau$ , then  $g'(t) = f(t)$  and  $g(0) = 0$ , implying

$$G(s) = F(s)/s.$$

( $\int_0^t \Rightarrow$  division by  $s$ )

# Ramp and impulse

- By integrating over  $t$  on  $\mathcal{L}[S(t)] = 1/s$ ,  
 $\mathcal{L}[t] = 1/s^2$  (ramp),  $\mathcal{L}[t^2/2] = 1/s^3$ ,  $\dots$   $\mathcal{L}[t^n/n!] = 1/s^{n+1}$
- By differentiating over  $t$  on  $\mathcal{L}[S(t)] = 1/s$ ,  
 $\mathcal{L}[S'(t)] = 1$   
... What is the derivative of a step?
- Dirac's  $\delta$  function (impulse):  $\delta(t) = S'(t)$



$$\delta(t) = 0 \text{ when } t \neq 0; \delta(t) = +\infty \text{ when } t = 0; \int_0^\infty \delta(t)dt = 1.$$

# Practice problems

- Exercise 1:  $\mathcal{L}[\sin(\omega t + \phi)] = ?$
- Exercise 2:  $\mathcal{L}[t^n e^{-at}] = ?$
- Exercise 3: If  $\mathcal{L}[f] = F(s)$ , then  $\mathcal{L}[f(t - \theta)S(t - \theta)] = ?$ 
  - ▶ Answer:  $e^{-\theta s}F(s)$
  - ▶ Delay by  $\theta \Rightarrow$  Multiplication by a factor of  $e^{-\theta s}$



# Solving ODE by Laplace transform

## Solve ODE [Seborg Ex.3.16]

Determine solution  $y(t)$  for ODE

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 9y = \cos t$$

with initial condition  $y(0) = 1, y'(0) = 2$ .

- Laplace transform on both sides:

$$[s^2Y(s) - sy(0) - y'(0)] + 6[sY(s) - y(0)] + 9Y(s) = \frac{s}{s^2 + 1}.$$

$$(s^2 + 6s + 9)Y(s) - s - 8 = \frac{s}{s^2 + 1} \quad \Rightarrow \quad Y(s) = \frac{s + (s + 8)(s^2 + 1)}{(s^2 + 6s + 9)(s^2 + 1)}$$

- Only have to find whose Laplace transform this is.

# Partial Fraction Expansion

- Goal: Invert the Laplace transform for any **fractional function of  $s$**

## Case 1: Denominator has simple roots

**Question:** Find  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+8} \right]$ .

- ① Find the roots:  $s^2 + 6s + 8 = (s + 2)(s + 4)$  (roots are  $-2, -4$ )

- ② Let

$$\frac{2s + 7}{s^2 + 6s + 8} = \frac{c_1}{s + 2} + \frac{c_2}{s + 4}$$

- ③ Determine  $c_1$  and  $c_2$  (by multiplying both sides by the denominator):

$$2s+7 = c_1(s+4)+c_2(s+2) \Rightarrow \begin{cases} c_1 + c_2 = 2 \\ 4c_1 + 2c_2 = 7 \end{cases} \Rightarrow \begin{cases} c_1 = 1.5 \\ c_2 = 0.5 \end{cases}$$

- ④ Hence,  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+8} \right] = \mathcal{L}^{-1} \left[ \frac{1.5}{s+2} + \frac{0.5}{s+4} \right] = 1.5e^{-2t} + 0.5e^{-4t}$ .

# Partial Fraction Expansion

## Case 2: Denominator has repeated roots

**Question:** Find  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+9} \right]$ .

① Find roots:  $s^2 + 6s + 9 = (s + 3)^2$  (roots are  $-3$  with multiplicity 2)

② Let

$$\frac{2s+7}{s^2+6s+9} = \frac{c_1}{s+3} + \frac{c_2}{(s+3)^2}$$

③ Determine  $c_1$  and  $c_2$ :

$$2s+7 = c_1(s+3) + c_2 \Rightarrow \begin{cases} c_1 = 2 \\ c_2 = 1 \end{cases}$$

④ Hence,  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+9} \right] = \mathcal{L}^{-1} \left[ \frac{2}{s+3} + \frac{1}{(s+3)^2} \right] = 2e^{-3t} + te^{-3t}$ .

# Partial Fraction Expansion

**Case 3:** Denominator has complex roots – Do it as usual.

**Question:** Find  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+13} \right]$ .

① Find the roots:  $s^2 + 6s + 13 = (s + 3 + 2i)(s + 3 - 2i)$

② Let

$$\frac{2s + 7}{s^2 + 6s + 13} = \frac{c_1}{s + 3 + 2i} + \frac{c_2}{s + 3 - 2i}$$

③ Determine  $c_1$  and  $c_2$  by  $2s + 7 = c_1(s + 3 - 2i) + c_2(s + 3 + 2i) \Rightarrow$

$$\begin{cases} c_1 + c_2 = 2 \\ (3 - 2i)c_1 + (3 + 2i)c_2 = 7 \end{cases} \Rightarrow \begin{cases} c_1 = 1 + 0.25i \\ c_2 = 1 - 0.25i \end{cases}$$

④ Hence,  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+13} \right] = (1 + 0.25i)e^{-3t-2it} + (1 - 0.25i)e^{-3t+2it}$ .

# Partial Fraction Expansion

**Case 3:** Denominator has complex roots – Do it as usual.

**Question:** Find  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+13} \right]$ .

⑤ Hence,  $\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+13} \right] = (1 + 0.25i)e^{-3t-2it} + (1 - 0.25i)e^{-3t+2it}$ .

⑥ In view of **Euler's formula**

$$e^{2it} + e^{-2it} = 2 \cos 2t, \quad e^{2it} - e^{-2it} = 2i \sin 2t,$$

we have

$$\mathcal{L}^{-1} \left[ \frac{2s+7}{s^2+6s+13} \right] = e^{-3t} [2 \cos 2t - 0.5 \sin 2t].$$

# Solving ODE by Laplace transform

## Solve ODE [Seborg Ex.3.16]

Determine solution  $y(t)$  for ODE  $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 9y = \cos t$  with initial condition  $y(0) = 1$ ,  $y'(0) = 2$ .

- From Laplace transform,

$$Y(s) = \frac{s + (s+8)(s^2+1)}{(s^2+6s+9)(s^2+1)} = \frac{s^3 + 8s^2 + 2s + 8}{(s+3)^2(s+i)(s-i)}$$

- Now do PFE. Let

$$\begin{aligned} \frac{s^3 + 8s^2 + 2s + 8}{(s+3)^2(s+i)(s-i)} &= \frac{c_1}{s+3} + \frac{c_2}{(s+3)^2} + \frac{c_3}{s+i} + \frac{c_4}{s-i} \\ s^3 + 8s^2 + 2s + 8 &= c_1(s+3)(s+i)(s-i) + c_2(s+i)(s-i) \\ &\quad + c_3(s+3)^2(s-i) + c_4(s+3)^2(s+i) \end{aligned}$$

# Solving ODE by Laplace transform

- Usual approach: Expand R.H.S. and match the coefficients
- \*\* Tricky approach: Match by plugging in the roots

$$s^3 + 8s^2 + 2s + 8 = c_1(s+3)(s+i)(s-i) + c_2(s+i)(s-i) + c_3(s+3)^2(s-i) + c_4(s+3)^2(s+i)$$

- ▶ Examine  $s = -3$ :  $47 = c_2(-3+i)(-3-i) \dots c_2 = 4.7$
- ▶ Examine  $s = -i$ :  $-i = c_3(-i+3)^2(-2i) \dots c_3 = 0.04 + 0.03i$
- ▶ Examine  $s = -i$ :  $i = c_4(i+3)^2(2i) \dots c_4 = 0.04 - 0.03i$
- ▶ Need another eq.:  $1 = c_1 + c_3 + c_4$  for the coeff. of  $s^3 \dots c_1 = 0.92$

- Finally,

$$\begin{aligned} y(t) &= \mathcal{L}^{-1} \left[ \frac{0.92}{s+3} + \frac{4.7}{(s+3)^2} + \frac{0.04 + 0.03i}{s+i} + \frac{0.04 - 0.03i}{s-i} \right] \\ &= 0.92e^{-3t} + 4.7te^{-3t} + (0.04 + 0.03i)e^{-it} + (0.04 - 0.03i)e^{it} \\ &= 0.92e^{-3t} + 4.7te^{-3t} + 0.08 \cos t + 0.06 \sin t \end{aligned}$$